

Adriana-Stefania Ciupeanu

Statistician, Scientific Machine-Learning Researcher, and Mathematical Modeller

 [cadrianas.github.io](https://github.com/cadrianas)  [cadrianas](#)

Profile

Bayesian inference, neural surrogate modelling, network science, and reproducible software for complex biological and health systems.

Experience

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| University of Manitoba , Postdoctoral Fellow, Department of Mathematics | Winnipeg, MB, Canada |
| <ul style="list-style-type: none">Lead the machine-learning extension of a global metapopulation model for forecasting pathogen-importation risk across 238 countries and territories.Conceived a mechanistically informed neural-surrogate methodology for likelihood-free Bayesian inference and probabilistic forecasting.Designed temporally conditioned neural surrogates and comparative graph-attention and recurrent models using transportation-network and location-level features.Built an end-to-end Python pipeline for simulation, sampling, PyTorch training, SMC-ABC inference, uncertainty propagation, validation, and benchmarking. | May 2026 – Oct 2026
6 months |
| University of Manitoba , Instructor, Mathematical Skills (MSKL 0100) | Winnipeg, MB, Canada |
| <ul style="list-style-type: none">Held full responsibility for course delivery, assessment design, grading, learning materials, feedback, and student support.Nominated for the Extended Education Instructor Excellence Award in 2026. | Sept 2025 – Dec 2025
4 months |
| University of Manitoba , Lead Teaching Assistant and Guest Lecturer, Mathematics of Data Science | Winnipeg, MB, Canada |
| <ul style="list-style-type: none">Delivered guest lectures on graph theory and network science and developed the associated teaching materials.Led grading of programming assignments and contributed datasets and revisions to course assessments. | 2023 – 2025
2 years |
| University of Manitoba , Undergraduate Research Mentor | Winnipeg, MB, Canada |
| <ul style="list-style-type: none">Guide development of a hybrid gravity-graph neural network model of inter-city mobility for spatial epidemic modelling.Provide methodological direction in graph neural networks, mobility modelling, experimental design, validation, and research communication. | 2026 – present
1 year |

Education

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| PhD University of Manitoba , Interdisciplinary Individual Studies - Mathematics and Statistics | Winnipeg, MB, Canada |
| <ul style="list-style-type: none">Supervisors: Dr. Julien Arino and Dr. Saman MuthukumaranaDissertation: Mathematical and statistical modelling during a pandemic event | 2019 – 2026 |
| MSc University of Manitoba , Mathematics | Winnipeg, MB, Canada |
| | 2016 – 2020 |
| MSc Queen's University , Mathematics and Statistics | Kingston, ON, Canada |
| | 2015 – 2016 |
| BSc Oxford Brookes University , Mathematics | Oxford, UK |
| | 2012 – 2015 |

Publications

- Mathematical Modeling of the Dynamics of COVID-19 Variants of Concern: Asymptotic and Finite-Time Perspectives** 2022
Adriana-Stefania Ciupeanu, Marie Varughese, Weston C. Roda, Donglin Han, Qun Cheng, Michael Y. Li
doi.org/10.1016/j.idm.2022.08.006 (Infectious Disease Modelling)
- Investigating Air Travel Network Changes in Canada, USA and Europe during COVID-19 Using Open Source Data** 2023
Adriana-Stefania Ciupeanu, Ashani Wickramasinghe, Saman Muthukumarana, Julien Arino
doi.org/10.21203/rs.3.rs-2426208/v1 (Research Square (Preprint))
- Preventing HPV-Induced Cervical Cancer in Alberta, Canada** 2026
Tasfia Akhter, Adriana-Stefania Ciupeanu, Weston C. Roda, Michael Y. Li, Sheila S. Plitt, Caitlin L. Charlton
(Submitted to the Canadian Journal of Public Health)

Skills

Programming and scientific computing: Python, R, Git, GitHub, LaTeX, PyTorch, PyTorch Geometric, NumPy, SciPy, pandas, scikit-learn

Bayesian and statistical methods: MCMC, SMC-ABC, parameter estimation, identifiability analysis, uncertainty quantification, model comparison, PRCC, eFAST

Machine learning: graph attention networks, LSTM and GRU models, neural surrogate modelling, mechanistically informed machine learning

Mathematical and network modelling: ODE models, metapopulation models, temporal and spatial networks, NetworkX, igraph

Research software: Python package development, automated testing, documentation, version control, reproducible computational workflows

Awards

- Extended Education Instructor Excellence Award nomination** 2026
University of Manitoba
- Faculty of Graduate Studies Travel Award** 2022-2024
University of Manitoba
- IMA Prize for Academic Excellence in Mathematics** 2015
Oxford Brookes University

Service

- Faculty Hiring Committee:** Graduate student representative, Department of Mathematics, University of Manitoba (2024-2025)
- Department and Graduate Student Association:** Graduate student representative, committee member, and department councillor, University of Manitoba (2018-2019)